# Nagoor Veeraboina

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## Summary

Software engineer with hands-on experience in **Python, SQL, backend development, machine learning, and LLM applications**, with enterprise software experience and an **IEEE-published Android malware study**. Experienced in **API development, data validation, debugging, database workflows, machine learning, and AI applications**.

## Technical Skills

**Programming:** Python, SQL
**Data Science:** EDA, Data Preprocessing, Feature Engineering, Model Evaluation
**Machine Learning:** scikit-learn, pandas, NumPy, TensorFlow, XGBoost, LightGBM
**Generative AI:** LLM Integration, RAG, Prompt Engineering

## Projects

**Logic Lens - AI Code Review Platform** Dec 2025 - Jun 2026
Python, FastAPI, React, PostgreSQL/pgvector, Groq API Live Demo | GitHub

- Built and deployed an **LLM-powered code review platform** supporting 4 workflows: issue detection, progressive DSA hints, time and space complexity analysis, and corrected-code generation with structured diffs.
- Designed the **FastAPI backend workflow** with request validation, task-specific prompting, response parsing, PostgreSQL persistence, JWT authentication, saved review history, and structured error handling.
- Implemented **RAG** across PDF, Markdown, and TXT documents using chunking, OpenAI embeddings, and pgvector similarity search to ground reviews and expose retrieved sources.
- Added **response streaming, runtime caching, rate limiting, and pytest-based API/unit testing** to improve reliability and control AI-backed API usage.

**CommuteCircle - Privacy-Aware Carpooling** Jun 2026 - Present
React Native (Expo), Python, FastAPI, PostgreSQL

- Developing a **mobile-first carpooling application** for listing rides and discovering trips with compatible routes and available seats.
- Built the application foundation and **ride-listing workflow** with origin, destination, departure time, and seat availability backed by PostgreSQL.
- Designed the backend and data flow to support route-aware ride discovery while keeping the application structure modular for future matching features.
- Developing **privacy-aware route matching** around public pickup points, walking-distance limits, and detour constraints to reduce disclosure of users' home addresses; matching remains in development.

## Professional Experience

**Associate Systems Engineer | Tata Consultancy Services** Aug 2024 - Nov 2025
Hyderabad, India | C#, .NET, Angular, SQL Server, Jira

- Built and maintained C#/.NET backend features for an automotive application used across European markets, writing SQL Server queries and stored procedures and tracking work in Jira.
- Analyzed client issues from Jira across frontend, backend, and database layers, identified root causes, and implemented required code and database updates.
- Shipped approved changes through the team's CI/CD deployment pipeline and supported application releases.

## Research Experience

**Android Malware Detection | IBM Security Collaboration** Aug 2023 - Jun 2024
Python, TensorFlow, CICMalDroid 2020, Bi-LSTM, LSTM-CNN, BiLSTM-CNN, SHAP, LIME

- Analyzed the **CICMalDroid 2020 dataset** containing **11,598 Android applications, 471 behavioral features, and 5 classes** covering benign, adware, banking, SMS malware, and riskware.
- Performed **EDA, data preprocessing, outlier analysis, class-distribution analysis, and feature selection**; selected the **top 50 features** using Random Forest feature importance.
- Addressed class imbalance with **SMOTE**, applied cross-validation and **10-trial random-search hyperparameter tuning**, and evaluated models using accuracy, precision, recall, F1-score, ROC-AUC, and MCC.
- Evaluated **3 deep-learning architectures - Bi-LSTM, LSTM-CNN, and BiLSTM-CNN** - achieving approximately **87.9% precision**; applied SHAP, LIME, and partial-dependence analysis for model interpretation.

**Publication:** *Towards Explainability Using ML and Deep Learning Models for Malware Threat Detection* - **IEEE AIIoT 2024** | **IEEE Xplore**

## Education

**Amrita Vishwa Vidyapeetham, Coimbatore** 2020 - 2024
Bachelor of Technology in Computer Science and Engineering
**Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Data Science, Machine Learning, Data Mining.

## Certifications

**AMD AI Academy:** AI Agents 101: Building AI Agents with MCP and Open-Source Inference Jul 2026
**Introduction to Big Data with Spark and Hadoop** - Coursera Nov 2023